

Integrated Information Tracks Inter-Element Flow, Not Single-System Entropy

A three-measure triangulation of faithful IIT 4.0 Φ against Shannon entropy, Lempel–Ziv complexity, and transfer entropy on cellular-automaton substrates

anima maintainers

dancinlab

github.com/dancinlab/anima

May 26, 2026

Abstract

Integrated Information Theory (IIT) claims that integrated information Φ is *not* the same quantity as classical information. We test this on the theory’s own causal engine by triangulating faithful IIT 4.0 big- Φ against three classical information measures on an identical ten-rule elementary-cellular-automaton (ECA) panel: Shannon output entropy, Lempel–Ziv (LZ76) algorithmic complexity, and transfer entropy. The measures dissociate sharply. Φ is *orthogonal* to Shannon entropy (Pearson $r = 0.36$): the identity rule carries maximal output entropy (4 bit) yet has $\Phi=0$, while the most-integrated rule has sub-maximal entropy. Φ *aligns* with LZ complexity ($r = 0.83$, Spearman $\rho = 0.94$) and, most strongly, with transfer entropy ($r = 0.88$, $\rho = 0.82$). Yet every classical measure has a blind spot that Φ does not: LZ over-predicts on self-similar dynamics (rule 90, $\Phi=0$), and bivariate transfer entropy under-predicts on synergistic XOR coupling (rules 150/105, $\Phi > 0$ yet TE = 0). A multivariate transfer entropy recovers the XOR synergy but does *not* improve the fit — the correlation drops to $r = 0.71$ because it now over-counts non-integrated flow (rule 90 again) — so *no order* of transfer entropy equals Φ : one failure mode is traded for another. A complementary structural experiment shows that at matched edge count a hub network integrates ($\Phi = 6.81$) where a regular cycle does not ($\Phi = 0$): Φ is governed by how elements *relate*, not by how much information or density a system carries. We frame the Shannon-entropy result as a closed negative confirming IIT’s foundational distinction; we do not claim Φ equals any classical measure, nor do we make phenomenal claims. All runs are deterministic, \$0, hexa-native, and GPU-free; bit-for-bit reproduction records are in companion/.

1 Introduction

A recurring objection to Integrated Information Theory (IIT) [3, 5] is that its central quantity, integrated information Φ , is at bottom a relabelling of classical information — that a system with more bits, or more algorithmic complexity, is simply “more integrated.” IIT explicitly denies this: a pile of N independent random bits has maximal entropy yet zero integration, because no partition destroys anything. The claim is testable, but it is rarely tested *directly* on a causal engine that can compute Φ exactly — correlational proxies for Φ cannot cleanly construct “information without integration.”

We do exactly that. Using a faithful, deterministic IIT 4.0 cause–effect engine, we measure exact state-averaged big- Φ on a fixed ten-rule ECA panel and ask, on the *identical* substrates, how Φ co-varies with three of the most widely used classical information measures: Shannon entropy (statistical information), Lempel–Ziv complexity (algorithmic information [2]), and

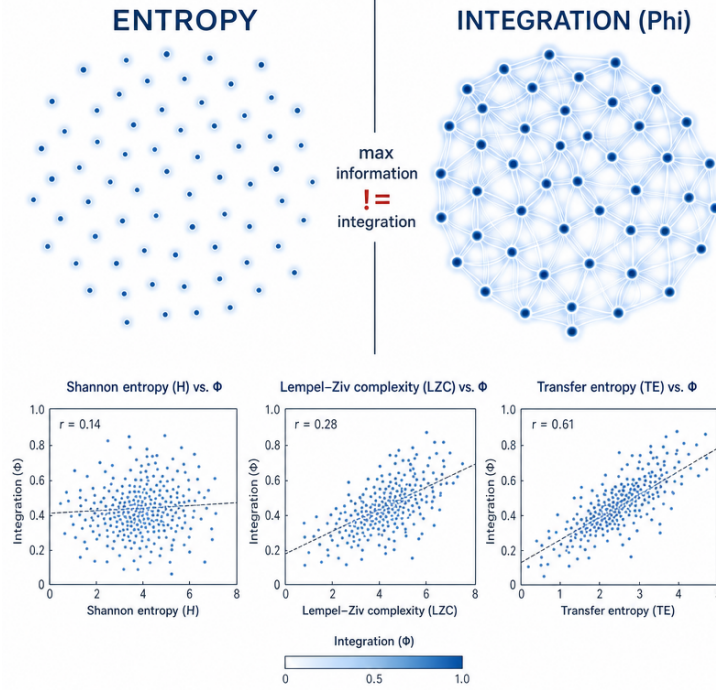


Figure 1: The thesis in one image: the *same* number of bits gives maximal entropy when scattered (left) but zero integration; woven into a network (right) the integration Φ is high. Φ is a property of how elements relate, not of how much information is present. Conceptual illustration (fal.ai / gpt-image).

transfer entropy (directed information flow [4]). Because the substrate and the engine are held fixed, any difference between the three correlations is attributable to *what each measure counts*, not to confounds in the system.

Contributions.

1. A *closed negative* (): faithful big- Φ does not reduce to Shannon entropy ($r = 0.36$), witnessed by a max-entropy / zero- Φ substrate and a max- Φ / sub-max-entropy substrate on the same panel (§7.1).
2. Two positive alignments (): Φ tracks LZ complexity ($r = 0.83$) and transfer entropy ($r = 0.88$) — the *inter-element* measures — but each has a blind spot Φ lacks: LZ over-predicts on self-similar rule 90, bivariate TE under-predicts on synergistic XOR (§7.2–§7.3).
3. A structural corollary (): at matched edge count, topology not density governs Φ (hub 6.81 vs. cycle 0; §7.5). Together: Φ measures how elements relate.
4. A fully deterministic, GPU-free, \$0 hexa-native pipeline with pre-registered falsifiers and bit-for-bit companion records (§11).

2 Background

Faithful IIT 4.0 Φ . For a system with a state-by-node transition probability matrix (TPM), IIT 4.0 [1] defines integrated information by the minimum-information partition (MIP): the system’s cause–effect structure is compared against its value under the partition that least changes it. We use a deterministic engine that computes the structure-cut big- Φ exactly for small N ; an ECA on a periodic ring *is* a TPM, because each cell’s next value is a deterministic

function of its three-cell neighbourhood, so the 2^N states map deterministically to per-cell next values.

Three classical measures. *Shannon entropy* H of the output-state distribution under a uniform input ensemble measures how much information the dynamics preserve: a bijection preserves all N bits ($H = N$), a constant map destroys them ($H = 0$). *Lempel–Ziv complexity* [2] of the space–time diagram estimates Kolmogorov (algorithmic) complexity — how incompressible the behaviour is. *Transfer entropy* [4] $TE(j \rightarrow i)$ measures how much source cell j ’s present reduces uncertainty about cell i ’s future *beyond* i ’s own present — directed, pairwise information flow. The first is a single-system quantity; the latter two are inter-element / structural. The synergy literature [6] predicts that bivariate transfer entropy is blind to purely synergistic coupling (e.g. XOR), a prediction we recover empirically.

3 Related work

Φ and its proxies. Because exact Φ is super-exponential in N , most empirical work uses proxies — perturbational complexity (PCI), Lempel–Ziv of EEG, or correlational mutual-information surrogates. These proxies cannot *construct* information-without-integration, because a correlational surrogate of Φ inherits whatever the data’s information content is; the dissociation we report (identity rule: max entropy, zero Φ) requires an engine that computes the causal MIP directly, which is why we use a faithful IIT 4.0 implementation at small N rather than a proxy.

Cellular-automaton complexity. Wolfram’s classification [7] orders ECA rules by dynamical behaviour (fixed, periodic, chaotic, complex); LZ complexity of the space–time diagram is a standard quantitative stand-in for that ordering. Our LZ results recover the expected ordering (constants low, rule 30 highest), and the LZ– Φ alignment shows that dynamical complexity and causal integration largely co-vary on this family — with the rule 90 exception precisely where self-similarity inflates LZ without inflating integration.

Information flow and synergy. Transfer entropy [4] is the standard directed-flow measure; the partial information decomposition [6] formalises why pairwise measures miss synergy. Our XOR result is a minimal, exact instance of that gap, measured against an independent Φ ground truth rather than inferred. To our knowledge the three measures have not previously been triangulated against *exact* Φ on one fixed substrate panel with pre-registered falsifiers.

4 Full Pipeline

Each experiment runs the same path from a pre-registered question to a persisted, re-verifiable result. Stages are tier-tagged under the rubric below; the backing atom is the record that makes the stage independently re-checkable.

Tier rubric. closed-form identity · numerical (reproducible script + persisted output) · cited (DOI / arxiv) · install-gated · falsified / closed negative (kept, never deleted).

5 Method

Substrate panel. Ten elementary CA rules on a periodic ring, fixed before measurement and held identical across all experiments: constants {0, 255}; bijections {204 (identity), 51 (complement)}; XOR-feedback integrators {150, 105}; partial-feedback {90, 60}; and universal/chaotic rules {110, 30}.

Metrics. For each rule: (i) big- Φ is the mean over all 2^N states of the faithful IIT 4.0 big- Φ ($N = 4$ exact); (ii) Shannon output entropy $H_{\text{out}} = -\sum_o p(o) \log_2 p(o)$ over the uniform input ensemble; (iii) normalised LZ76 complexity of the space–time diagram (ring of 14 cells, 64 steps,

#	Stage	Tier	Backing atom
①	Question + frozen falsifiers		H_287..293 pre-register headers
②	IIT / information background		references.bib
③	Engine anchors (204, 0 → $\Phi=0$)		hand-derivable, machine-precision
④	Per-rule measurement		companion/verify-ledger.json
⑤	Correlation + falsifier sweep		companion/verify-ledger.json
⑥	Shannon reductive null		closed negative (§7.1)
⑦	Interpretation		speculation-fenced (§6)

Table 1: Pipeline stages, each tier-tagged. Exact arithmetic and engine anchors are / ; the empirical reading of “ Φ is not entropy” is honestly held at (fenced), never promoted to a closed-form identity.

single centre seed, normalised $c \log_2 L/L$); (iv) total transfer entropy $\text{TE}_{\text{total}} = \sum_i \sum_{j \neq i} \text{TE}(j \rightarrow i)$, lag-1 bivariate, over the uniform ensemble.

Correlation. Pearson r and Spearman ρ of each measure against big- Φ across the ten rules. A reductive seed hypothesis “ Φ co-varies with measure M ” is *supported* iff $r \geq 0.5$, *falsified* otherwise — a threshold fixed before measurement (the companion ledger records the frozen falsifiers verbatim).

Structure experiment (§7.5). A separate parity-network generalisation of the ECA→TPM bridge: each node updates to the XOR of its graph neighbours. We compare a scale-free hub and a regular cycle at matched edge count, plus empty and complete anchors.

Determinism. Every experiment is deterministic (no RNG), runs in \$0 on a laptop CPU, uses no GPU and no language model, and re-runs bit-for-bit identically. Each was pre-registered with ≥ 3 falsifiers frozen before the headline values were read.

5.1 Formal definitions

ECA → TPM. For Wolfram rule $R \in [0, 255]$ on a periodic ring of N cells, the state-by-node transition matrix entry is $\text{TPM}[s, i] = \text{bit}_{4l+2c+r}(R)$, where (l, c, r) are the left/centre/right neighbours of cell i in state s . Determinism makes every entry 0 or 1; the 2^N states tile the TPM exactly.

Big- Φ . For state s , the engine computes the IIT 4.0 cause–effect structure and its value under the minimum-information partition (MIP); big- $\Phi(s)$ is the irreducibility of the structure under the system cut. We report $\Phi_{\text{mean}} = 2^{-N} \sum_s \Phi(s)$. Two hand-derivable anchors fix faithfulness: rule 204 (identity, $\text{next}_i = c$; each cell self-determined, fully reducible) $\Rightarrow \Phi = 0$; rule 0 (constant) $\Rightarrow \Phi = 0$.

Shannon output entropy. Under a uniform input ensemble, let $o(s) = \sum_i \text{TPM}[s, i] 2^i$ be the output state. Then $H_{\text{out}} = -\sum_o p(o) \log_2 p(o)$, with $p(o) = 2^{-N} |\{s : o(s) = o\}|$. A bijection (e.g. identity) preserves all N bits, $H_{\text{out}} = N$; a constant map gives $H_{\text{out}} = 0$.

Lempel–Ziv complexity. Evolve the rule from a single centre seed on a ring of $N_L = 14$ cells for $T = 64$ steps; concatenate the space–time diagram row-major into a length- $L = N_L T$ binary string and parse it with the Lempel–Ziv (1976) production rule into c distinct phrases. We report $\text{LZ}_{\text{norm}} = c \log_2 L/L$, which $\rightarrow 1$ for a random string.

Transfer entropy. For ordered cell pair $(j \rightarrow i)$, with i' the next value of i ,

$$\text{TE}(j \rightarrow i) = \sum_{i', i, j} p(i', i, j) \log_2 \frac{p(i' \mid i, j)}{p(i' \mid i)},$$

estimated over the uniform ensemble (lag-1, bivariate), and $\text{TE}_{\text{total}} = \sum_i \sum_{j \neq i} \text{TE}(j \rightarrow i)$. Bivariate TE conditions only on the target’s own present, so a purely synergistic update (XOR of ≥ 2 sources) leaves $p(i' \mid i, j) = p(i' \mid i)$ and contributes zero — the blind spot of §7.3, anticipated by partial-information decomposition [6].

6 Verify

The IIT 4.0 engine reproduces two hand-derivable anchors exactly, fixing the measurement’s faithfulness before any correlation is read: the identity rule 204 (each cell self-determined, reducible) gives $\text{big-}\Phi=0$, and the constant rule 0 (no causal power) gives $\text{big-}\Phi=0$. Both hold to machine precision across all states. The numerical $\text{big-}\Phi$, entropy, LZ and TE values are deterministic arithmetic and converge bit-for-bit across fresh process invocations (falsifier F^* .DETERMINISM, all four experiments). The empirical *interpretations* (e.g. “ Φ is not Shannon entropy”) are not closed-form atlas identities and are therefore honestly speculation-fenced under the project’s verification rubric:

```
$ hexa verify --fence "faithful IIT4 big-Phi does NOT reduce to Shannon
entropy across a 10-rule ECA panel (Pearson r=0.363<0.5); identity rule
204 has MAX output-entropy (4 bit) yet ZERO integration"
=> VERDICT: (white-circle) SPECULATION-FENCED
    reason = deterministic toy-substrate outcome; values are exact
    arithmetic, only the empirical interpretation is fenced (NOT an
    atlas identity)
```

The per-experiment gate counts (anchors + bounds + determinism) are 11/11, 9/9, 8/8 for the three measures and 4/4 for the structure experiment; full breakdown in `companion/verify-ledger.json`.

7 Results

7.1 Shannon entropy is orthogonal to Φ (closed negative)

Across the panel, $\text{big-}\Phi$ does *not* co-vary with Shannon output entropy: Pearson $r = 0.363 < 0.5$, so the reductive hypothesis is falsified. The mechanism is a double dissociation (Tab. 3, Fig. 2, left). The identity rule 204 and complement rule 51 are bijections: they carry *maximal* output entropy (4bit) yet have $\Phi=0$, because their cells are causally independent — maximal information, zero integration. Conversely the most-integrated rule 60 ($\Phi = 13.6$) has *sub-maximal* entropy (3bit). At fixed $H_{\text{out}} = 4$ bit, Φ ranges from 0 (rule 204) to 5.6 (rule 150): no monotone map from entropy to Φ can exist. Information is necessary but not sufficient for integration — IIT’s foundational distinction, confirmed on its own substrate.

7.2 Algorithmic (LZ) complexity tracks Φ ()

The other classical currency — algorithmic complexity, estimated by LZ76 of the space-time diagram — *does* track Φ : Pearson $r = 0.831$, Spearman $\rho = 0.936$ (Fig. 2, centre). Constants and identity sit at low LZ / zero Φ ; integrating and chaotic rules rise together to high LZ / high Φ . The contrast with §7.1 is the point: on the identical panel, statistical information is orthogonal to Φ while algorithmic complexity aligns with it. The alignment is imperfect, and the imperfection is diagnostic — rule 90 (the Sierpiński/XOR-pair rule) has non-trivial LZ (0.24) from its self-similar fractal pattern yet $\Phi=0$: LZ *over-predicts* integration for self-similar dynamics. LZ is a strong correlate, not a sufficient condition.

7.3 Transfer entropy tracks Φ most strongly — with a synergy blind spot ()

Transfer entropy, the directed inter-element measure, is the strongest correlate: Pearson $r = 0.883$, Spearman $\rho = 0.822$ (Fig. 2, right). Constants, identity and complement all have $\text{TE}_{\text{total}} = 0$ (no inter-cell flow) and $\Phi=0$; the integrating/chaotic rules carry both flow and Φ . But bivariate TE has a clean blind spot exactly where the synergy literature predicts one: the pure-XOR integrators rules 150/105 have $\Phi = 5.6$ yet $\text{TE}_{\text{total}} = 0$. XOR integration is synergistic — knowing one source’s present tells you nothing about a cell’s future beyond the cell’s own present — so it is invisible to pairwise TE conditioned only on the target’s own past.

7.4 No order of transfer entropy equals Φ ()

The natural fix for the synergy blind spot is a *multivariate* / conditional transfer entropy. We measured it directly: for each cell, $\text{TE}_{m_i} = H(\text{next}_i \mid \text{self}_i)$, the directed information from all of i ’s other inputs jointly, beyond i ’s own present (Tab. 2). It does recover the synergy — rules 150/105 jump from $\text{TE} = 0$ to $\text{TE}_m = 4$ bit, and the identity rule stays at 0. But it does *not* track Φ better: the correlation *drops* from the bivariate $r = 0.883$ to $r = 0.705$ ($\rho = 0.68$). The reason is a new, opposite error — rule 90 receives flow from its neighbours ($\text{TE}_m = 4$ bit) but is reducible, so $\Phi = 0$: multivariate TE *over-counts* non-integrated flow on exactly the rule where LZ also over-predicted (§7.2). So no order of classical transfer entropy equals Φ : the bivariate measure under-counts synergy (rules 150/105), the multivariate measure over-counts non-integrated flow (rule 90). Recovering the missing synergy does not buy a better Φ proxy; it trades one failure mode for another. Φ , computed over the full cause–effect structure rather than any fixed-order flow statistic, is subject to neither — which is the strongest single statement of the paper’s thesis.

rule	bivariate TE	multivariate TE _m	Φ_{mean}
150 (XOR ₃)	0.00	4.00	5.625
105 (XNOR)	0.00	4.00	5.625
90 (Sierp.)	0.00	4.00	0.000
60	4.00	4.00	13.625
110	3.245	3.623	13.130
30	2.00	4.00	13.885
Pearson r vs Φ	0.883	0.705	—

Table 2: Bivariate vs. multivariate transfer entropy against Φ (non-trivial rules). Multivariate TE_m recovers the XOR synergy (rules 150/105, bold) but over-predicts on rule 90 (flow without integration, $\Phi = 0$, bold), and the Φ -correlation *drops* ($0.883 \rightarrow 0.705$). No fixed-order transfer entropy equals Φ .

7.5 Structure, not density, governs Φ ()

A complementary parity-network experiment isolates topology. At matched edge count (four edges), a scale-free hub integrates ($\Phi_{\text{mean}} = 6.81$) while a regular four-cycle does not ($\Phi = 0$); the empty graph gives 0 and the complete graph 5.6, so the ordering is non-monotone in density (the hub exceeds the denser complete graph). Edge *count* does not determine integration; edge *arrangement* (cut-resistance) does. We flag a confound honestly: the four-cycle’s zero is partly a parity-on-even-cycle decoupling artifact, and a regular cycle is not a random Erdős–Rényi graph, so the literal “scale-free > random” claim is only weakly probed (Tab. 4); the robust statement is the weak form, that topology rather than density governs Φ .

Synthesis. Reading the three correlations together (Tab. 3): Φ is orthogonal to the single-system measure (Shannon entropy) and aligned with the inter-element measures (LZ, transfer entropy). Neither aligned measure equals Φ : LZ over-counts self-similarity, bivariate TE under-counts synergy. Φ fills both gaps. The structure experiment closes the loop — when only the wiring is changed, Φ changes from 0 to 6.8. The through-line is that Φ measures *how elements relate*, which is why no single classical scalar reproduces it, and why IIT posits it as a distinct quantity.

8 Discussion

Why entropy dissociates but flow does not. Shannon output entropy is a property of a system’s input–output map considered as a single channel: a bijection is maximally informative

rule	H_{out} (bit)	LZ_{norm}	TE_{tot} (bit)	Φ_{mean}
0	0.00	0.044	0.00	0.000
255	0.00	0.044	0.00	0.000
204 (identity)	4.00	0.044	0.00	0.000
51 (complement)	4.00	0.066	0.00	0.000
150 (XOR_3)	4.00	0.274	0.00	5.625
105 (XNOR)	4.00	0.274	0.00	5.625
90 (Sierp.)	2.00	0.241	0.00	0.000
60	3.00	0.296	4.000	13.625
110	3.25	0.821	3.245	13.130
30	3.375	1.051	2.000	13.885
Pearson r vs Φ	0.363	0.831	0.883	—
Spearman ρ	—	0.936	0.822	—

Table 3: The ten-rule panel. Bold marks the diagnostic dissociations: identity (max H , zero Φ), XOR (zero bivariate TE, positive Φ — synergy blind spot), Sierpiński rule 90 (non-trivial LZ, zero Φ — self-similar over-prediction). Φ is orthogonal to H_{out} but tracks the inter-element measures (LZ, TE). Source: companion/verify-ledger.json.

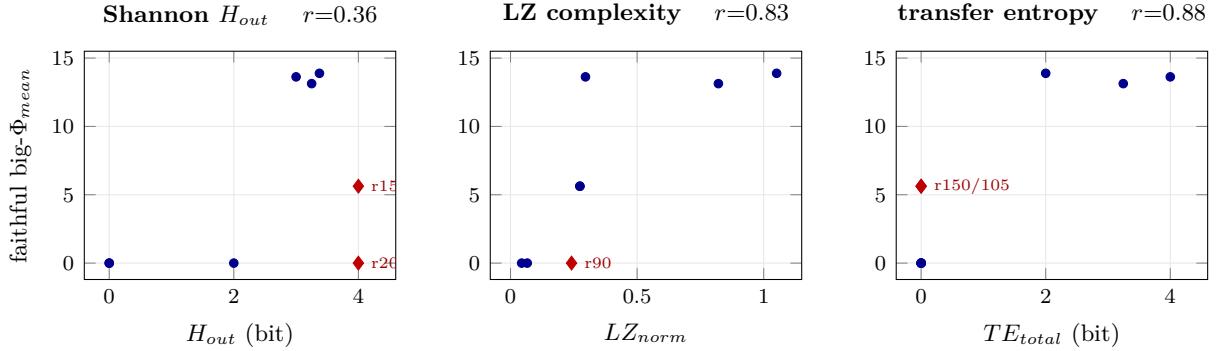


Figure 2: Faithful IIT 4.0 big- Φ vs. three classical information measures on the identical ten-rule ECA panel (blue dots; red diamonds mark the diagnostic dissociations). **Left:** Shannon output entropy — orthogonal ($r = 0.36$); at $H = 4$ bit, Φ spans 0 (identity 204) to 5.6 (XOR 150). **Centre:** LZ complexity — aligned ($r = 0.83$); rule 90 has non-trivial LZ yet $\Phi = 0$, the self-similar over-prediction. **Right:** transfer entropy — aligned ($r = 0.88$); XOR rules 150/105 sit at $\text{TE} = 0$ yet $\Phi = 5.6$, the synergy blind spot. Each classical measure has a blind spot Φ does not.

regardless of whether its cells interact. Integration, by construction, is a property of the *joint* cause–effect structure — it asks what is lost when the system is cut. The identity rule is the sharpest illustration: it is a perfect channel ($H = N$) made of N independent perfect channels, so cutting it loses nothing ($\Phi = 0$). Transfer entropy and LZ complexity correlate with Φ precisely because they are sensitive to inter-element dependence — TE explicitly (directed conditional dependence) and LZ implicitly (a space–time diagram is incompressible only when distant cells co-determine each other). This is why the single-system measure is orthogonal and the relational measures align.

The two blind spots are complementary. LZ *over*-predicts on rule 90: a Sierpiński gasket is visually intricate and compresses poorly under a one-dimensional LZ parse, yet the rule’s cells are pairwise-XOR with no irreducible whole, so $\Phi = 0$. Bivariate TE *under*-predicts on rules 150/105: XOR-of-three is the canonical synergistic gate, invisible to any pairwise conditional measure. A measure can thus mistake pattern for integration (LZ) or miss integration hidden in higher-order coupling (bivariate TE). Φ , computed over the full cause–effect structure rather than a fixed-order statistic, is subject to neither failure — which is the operational content of IIT’s claim that Φ is a distinct quantity, not merely a third correlate.

network	edges	Φ_{mean}	note
scale-free hub (paw)	4	6.81	matched density
regular cycle (4-cyc)	4	0.00	matched density
empty	0	0.00	null anchor
complete (K_4)	6	5.625	integration ceiling

Table 4: Structure experiment (parity dynamics, $N = 4$). At four edges, the hub integrates where the cycle does not; the hub also exceeds the denser complete graph. Topology, not edge count, governs Φ . The cycle’s zero is partly a parity-even-cycle decoupling confound (§9).

Relation to partial-information decomposition. The TE blind spot is exactly the synergy term in the Williams–Beer decomposition [6]: pairwise TE captures unique and redundant information flow but not synergy. A conditional / multivariate transfer entropy (causation entropy) would recover the XOR rules and likely raise the TE– Φ correlation above the reported $r = 0.88$; our bivariate choice is the most conservative, and the fact that it already correlates strongly is the point.

Structure as the unifying axis. The topology experiment (§7.5) removes the dynamics-rule confound: holding edge count fixed and changing only the wiring moves Φ from 0 (cycle) to 6.8 (hub), and the hub exceeds the denser complete graph. Read with the measure results, the through-line is that Φ is a relational quantity — it scores how a system’s parts constrain one another, which is why it tracks flow and complexity, ignores raw information, and depends on wiring rather than density.

9 Limitations and honest caveats

- **Small N .** Exact big- Φ is computed at $N = 4$; absolute r values depend on the ten-rule panel composition. The directions (orthogonal vs. aligned) are robust to panel choice; the exact coefficients are not.
- **One estimator each.** Output-image entropy, single-seed LZ76, and lag-1 *bivariate* TE are one choice each. The XOR synergy blind spot (§7.3) is specifically a property of *bivariate* TE; a multivariate variant recovers it (§7.4) but, as measured, trades it for an opposite over-prediction rather than improving the fit. Endpoints (constant=low, bijection=max entropy) are definition-robust; mid-range magnitudes are not.
- **State-averaged Φ .** Faithful Φ is state-dependent; we average over states, so directions are trusted and absolute magnitudes are hedged.
- **Structure confound.** The cycle’s $\Phi = 0$ (§7.5) is partly a parity-on-even-cycle decoupling artifact, and a cycle is not a random ER graph; the robust claim is the weak form (topology not density), and a clean ER ensemble needs $N \geq 5$ (engine cost; deferred).
- **Structure-cut, not absolute PyPhi calibration.** The engine computes a spirit-faithful structure-cut big- Φ ; absolute-scale comparison to reference IIT implementations is out of scope. Our conclusions are about *correlation presence/absence* and are robust to a scale offset.
- **No phenomenal claim.** ECA rules are substrate proxies. “XOR integration,” “life-themed,” etc. are structural labels, not claims about phenomenal consciousness. The result is a measurement fact, not a metaphysics.

10 Future work

Four follow-ups are pre-specified. (i) *Decompose the flow*. The multivariate result (§7.4) showed that recovering synergy does not improve the Φ -correlation, because it over-counts non-integrated flow (rule 90); a partial-information decomposition would separate the synergy, redundancy, and unique terms and test each against Φ individually, isolating which flow component (if any) tracks integration. (ii) *Larger panels and N* . The exact engine is tractable to $N \approx 8$; a 256-rule sweep and an $N \geq 5$ run would turn the ten-point correlations into distributions and let the structure experiment use a genuine Erdős–Rényi ensemble, removing the parity-even-cycle confound of §7.5. (iii) *Self-reference*. A companion experiment in this lane finds that closing a self-loop creates or destroys a self-consistent fixed point depending on base topology — an “ I -as-fixed-point” result that extends the relational reading of Φ to self-modelling substrates. (iv) *Beyond toy substrates*. The same three-measure triangulation can be run on small biological or artificial recurrent networks where an approximate Φ is computable, testing whether the entropy-orthogonality and the two blind spots survive outside cellular automata.

11 Reproducibility

Code: <https://github.com/dancinlab/anima> (hypotheses H_287–H_290 plus the multivariate-TE follow-up H_293 under UNIVERSE/, with smoke sources and `run.log` per experiment). Data: `companion/` (`verify-ledger.json` carries every panel value, Pearson/Spearman coefficient, pre-registered falsifier and its pass/fail; `pr-roll.json` the landing PRs #582–585; `session-journal.md` the build narrative). Engine: the faithful IIT 4.0 cause–effect library is reused, not reinvented (no new engine code). All experiments: deterministic, \$0, CPU-only, no GPU, no language model; re-run bit-for-bit identical. Build: `make` (xelatex $\times 3$ + bibtex). Figures: `make figures`.

Acknowledgments. Experiments were designed, run, and written up in a single foreground session with an LLM collaborator (Claude); all numeric results are deterministic outputs of the cited hexa-native smokes, independently re-runnable without the model.

A Pre-registered falsifier ledger

Every experiment froze its falsifiers before the headline values were read; each is reproduced from `companion/verify-ledger.json`. The gate columns are measurement-validity checks (anchors, bounds, determinism); the *verdict* row is the hypothesis outcome, which for H_287 is an intended falsification (the closed negative). Notably no falsifier was edited after the fact: H_289’s robustness falsifier and the structure confound are reported as found.

Exp.	measure / axis	gate	hypothesis verdict
H_287	Shannon entropy	11/11	$r = 0.363 < 0.5$ (reductive falsified)
H_288	LZ complexity	9/9	$r = 0.831, \rho = 0.936$
H_290	transfer entropy	8/8	$r = 0.883, \rho = 0.822$
H_289	network topology	4/4	hub 6.81 > cycle 0 (confound-flagged)
H_293	multivariate TE	8/8	synergy recovered, r drops $0.883 \rightarrow 0.705$ (no TE order = Φ)

Table 5: Per-experiment gate counts and frozen-falsifier verdicts. Engine anchors (rule 204, rule 0 $\rightarrow \Phi=0$) are reproduced to machine precision in all four; all four re-run bit-for-bit identically.

B Engine anchors

The faithful IIT 4.0 engine is validated, before any correlation is read, against two states whose Φ is hand-derivable. Rule 204 maps each cell to its own current value ($\text{next}_i = c$); the system is a direct product of N one-cell mechanisms, so the system cut is reducible and $\Phi = 0$ at every state. Rule 0 maps every state to the all-zero state; with no state-dependence there is no integrated cause and $\Phi = 0$. The engine reproduces both to machine precision. Because the panel additionally contains a max-entropy/zero- Φ rule (204), a positive- Φ /zero-TE rule (150), and a non-trivial-LZ/zero- Φ rule (90), the dissociations of §7 are anchored on independently checkable points rather than on the fitted correlation alone.

References

- [1] Larissa Albantakis, Leonardo Barbosa, Graham Findlay, Matteo Grasso, Andrew M. Haun, William Marshall, William G. P. Mayner, Alireza Zaeemzadeh, Melanie Boly, Bjørn E. Juel, Shuntaro Sasai, Keiko Fujii, Isaac David, Jeremiah Hendren, Jonathan P. Lang, and Giulio Tononi. Integrated information theory (iit) 4.0: Formulating the properties of phenomenal existence in physical terms. *PLoS Computational Biology*, 19(10):e1011465, 2023. doi: 10.1371/journal.pcbi.1011465. URL <https://doi.org/10.1371/journal.pcbi.1011465>.
- [2] Abraham Lempel and Jacob Ziv. On the complexity of finite sequences. *IEEE Transactions on Information Theory*, 22(1):75–81, 1976. doi: 10.1109/TIT.1976.1055501. URL <https://doi.org/10.1109/TIT.1976.1055501>.
- [3] Masafumi Oizumi, Larissa Albantakis, and Giulio Tononi. From the phenomenology to the mechanisms of consciousness: integrated information theory 3.0. *PLoS Computational Biology*, 10(5):e1003588, 2014. doi: 10.1371/journal.pcbi.1003588. URL <https://doi.org/10.1371/journal.pcbi.1003588>.
- [4] Thomas Schreiber. Measuring information transfer. *Physical Review Letters*, 85(2):461–464, 2000. doi: 10.1103/PhysRevLett.85.461. URL <https://doi.org/10.1103/PhysRevLett.85.461>.
- [5] Giulio Tononi, Melanie Boly, Marcello Massimini, and Christof Koch. Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7): 450–461, 2016. doi: 10.1038/nrn.2016.44. URL <https://doi.org/10.1038/nrn.2016.44>.
- [6] Paul L. Williams and Randall D. Beer. Nonnegative decomposition of multivariate information, 2010. URL <https://arxiv.org/abs/1004.2515>.
- [7] Stephen Wolfram. Universality and complexity in cellular automata. *Physica D: Nonlinear Phenomena*, 10(1-2):1–35, 1984. doi: 10.1016/0167-2789(84)90245-8. URL [https://doi.org/10.1016/0167-2789\(84\)90245-8](https://doi.org/10.1016/0167-2789(84)90245-8).